

Trigonometric Identities

Pythagorean Identity

$$\sin^2 x + \cos^2 x = 1$$

Double Angle Formulas

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

Power-Reducing Formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2}$$

Product Identity

$$\sin^2 x \cos^2 x = \frac{1}{4} \sin^2(2x)$$

Product-to-Sum Identities

$$\sin x \cos y = \frac{1}{2} [\sin(x + y) + \sin(x - y)]$$

$$\cos x \cos y = \frac{1}{2} [\cos(x + y) + \cos(x - y)]$$

$$\sin x \sin y = \frac{1}{2} [\cos(x - y) - \cos(x + y)]$$

Useful for Integration

To simplify integrals involving $\sin^2 x$, $\cos^2 x$, or their product, use:

- $\int \sin^2 x \, dx = \int \frac{1 - \cos(2x)}{2} \, dx$
- $\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx$
- $\int \sin^2 x \cos^2 x \, dx = \int \frac{1}{4} \sin^2(2x) \, dx$